

# ECE557 Lab 4 Report

## Square Wave Tracking for the Cart-Pendulum Robot

University of Toronto

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## 1 Introduction

The objective of this lab is to design and implement output feedback control strategies for a cart-pendulum robot system. The primary goal is to achieve asymptotic tracking of a square wave reference signal for the cart position, while maintaining the pendulum in the upright unstable equilibrium. Unlike prior experiments that relied on state feedback, this lab utilizes an observer to estimate the complete state vector from limited state measurements ( $z$  and  $\theta$ ). We evaluate the performance of a standard output feedback controller and an augmented integral controller through both MATLAB (Simulink) simulations and physical experimentation. Figure 1 outlines the block diagram and equations for the cart-pendulum robot system in this lab.

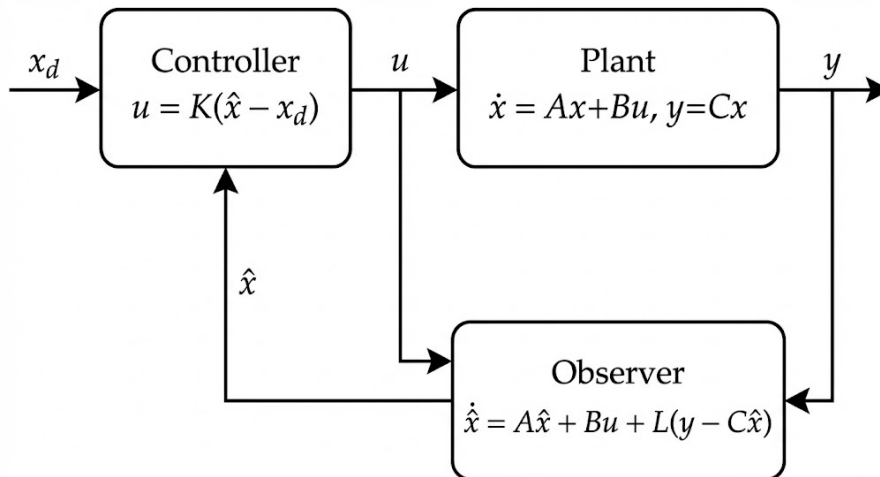


Figure 1: Block diagram and equations for the cart-pendulum system used in Lab 4

## 2 4.3: Simulation Comparison of State and Output Feedback Controllers

- 2.1 Provide one figure overlaying the signals  $(z(t), \theta(t), u(t))$ , corresponding to state and output feedback controllers when the eigenvalues of  $A+BK$  are at -5 and those of  $A-LC$  are at -10.

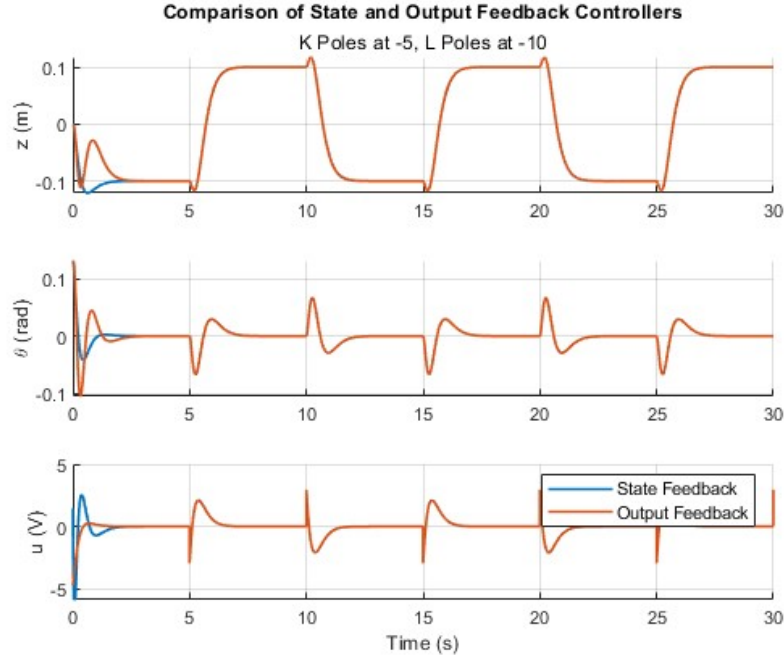


Figure 2: Plot of  $(z(t), \theta(t), u(t))$  when the eigenvalues of  $A+BK$  are at -5 and those of  $A-LC$  are at -10.

Figure 2 displays the simulation results for  $(z(t), \theta(t), u(t))$  between the state and output feedback for the cart-pendulum system tracking a square wave signal.

### 2.2 Discuss the differences between state feedback and output feedback solutions in the figures above

The primary difference between the state and output feedback solutions are only visible during the initial transient phase ( $t \leq 3s$ ), where the output feedback controller (orange) exhibits a sharper spike in control effort and angle deviation compared to the smoother response of the state feedback (blue) controller. This discrepancy arises due to the output feedback system relying on an observer that must first overcome initial estimation errors before it can accurately track the state. However, because the observer dynamics are design to be faster than the controller dynamics (i.e eigenvalues at -10 as opposed to -5), the estimation error decays rapidly. As a result of this, once the observer converges, the output feedback performance becomes identical to the ideal state feedback response.

**2.3 Provide a second figure when the eigenvalues of  $A - LC$  are moved to -40, and discuss the changes you observe. Print  $T_s$  and  $T_{sat}$  for this latter simulation**

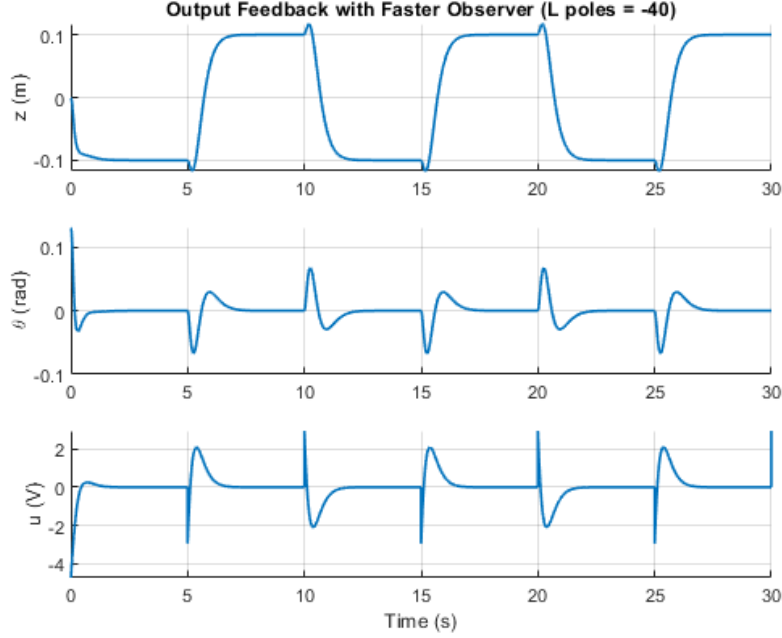


Figure 3: Plot of  $(z(t), \theta(t), u(t))$  when the eigenvalues of  $A-LC$  are moved to -40.

$$T_s = 1.6869 \text{ seconds } T_{sat} = 0 \text{ seconds}$$

When the observer eigenvalues are moved from -10 to -40, the system exhibits a significantly faster convergence of the state estimate to the true state. Figure 3 shows these results visually. The initial transient phase ( $t \leq 1s$ ), and the initial deviation in pendulum angle is much narrower in duration. This behavior aligns with the expected performance, the estimation error dynamics are governed by  $\dot{e} = (A - LC)e$ . With the eigenvalues moved to -40, the error decays according to  $e^{40t}$  much faster than the previous design of  $e^{10t}$ . Thus, the controller aligns with the state feedback behavior almost instantaneously.

**2.4 Discuss how you tuned  $K$  to meet the specifications. Provide the parameters for the tuning (either the eigenvalues or the  $Q, R$  matrices) and provide  $T_s$  for your output feedback controller**

To satisfy the performance specifications of  $T_s$  less than 1s, we tuned the state feedback gain  $K$  using eigenvalue assignment. The goal was to place the closed-loop poles sufficiently far enough left to accelerate the system's convergence rate while monitoring the trade-off with control effort saturation.

With the initial eigenvalues of  $(A+BK)$  at -5, the settling time resulted in 1.69 seconds. To decrease this, we shifted the poles to be more negative:  $eig(A+BK) = [-8, -8.01, -8.02, -8.03]$ . The observer eigenvalues were maintained at -40 as per the lab instructions. With the updated parameters, the output feedback controller achieved  $T_s = 0.88325s$  and  $T_{sat} = 9.1454s$ , achieving the  $T_s$  requirement but having a notably long saturation time.. The eigenvalues of  $(A+BK)$  were then moved to  $eig(A+BK) = [-7.5, -7.51, -7.52, -7.53]$ . These tuned parameters resulted in  $T_s = 0.98354s$  and  $T_{sat} = 5.1243s$ . These results are an optimal trade-off given the constraints.

We achieve the required  $T_s$  while greatly decreasing the  $T_{sat}$  compared to the initial update, avoiding "excessive" saturation.

### 3 5.1: Physical Experimentation of Output Feedback Controller

#### 3.1 Explain the trial and error procedure that you followed to get the best performance on the experimental setup, and provide the parameters of the controllers (either the eigenvalues or the Q, R matrices) that gave the best results.

The controller implemented on the Arduino was an LQR-based output-feedback controller designed using the linearized plant model. Initial attempts using pole placement (as designed in Section 4.3) consistently destabilized the physical system due to unmodeled dynamics, friction in the cart rail, and sensor quantization noise. The deterministic pole placement approach, while mathematically sound for the idealized model, proved too sensitive to real-world imperfections and lacked a systematic framework for balancing competing control objectives.

We transitioned to an LQR design approach, which provided a more principled tuning framework through the state-weighting matrix  $Q$  and control penalty  $R$ . The LQR cost function being minimized is:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt \quad (1)$$

where  $Q = \text{diag}(q_1, q_2, q_3, q_4)$  penalizes deviations in cart position, cart velocity, pendulum angle, and angular velocity respectively, while  $R$  penalizes control effort. The iterative tuning process followed a trial and error to finally arrive at:

#### Final Design (BEST PERFORMANCE)

- **Parameters:**  $Q = \text{diag}(0.5, 0, 150, 0)$ ,  $R = 0.01$
- **Resulting gain:**  $K = [7.0711, 21.334, -143.63, -13.307]$
- **Closed-loop eigenvalues:** Approximately  $\{-5.3, -6.9, -18.2 \pm 10.5i\}$
- **Result:**
  - Pendulum maintained near-vertical ( $|\theta| < 0.1$  rad) throughout square wave tracking
  - Cart position settled to within 2% of steady-state value
  - Control effort remained below saturation limit (5.875 V) except during brief transients

#### Observer Design

The observer design used eigenvalues placed at:

$$p_L = \{-40, -40.001, -40.002, -40.003\} \quad (2)$$

These fast observer poles (approximately 6-8 times faster than the slowest controller pole) ensure rapid state estimation convergence without excessive noise amplification. The resulting observer gain matrix is:

$$L = \begin{bmatrix} 71.187 & -0.00088 \\ 972.56 & 1.865 \\ -26.698 & 80.004 \\ -1900.5 & 1635.6 \end{bmatrix} \quad (3)$$

The large magnitudes in certain entries of  $L$  (particularly  $L_{42} = 1635.6$  and  $L_{41} = -1900.5$ ) reflect the high observer bandwidth required to accurately track the fast pendulum dynamics. The separation principle ensures that the observer dynamics do not interfere with the controller performance, provided the observer is sufficiently faster than the closed-loop system.

**3.2 Provide an experimental plot of the signals  $(z(t), \theta(t), u(t))$ , as well as the settling time. Discuss the results: whether they are satisfactory, whether they conform to your expectations, and so on.**

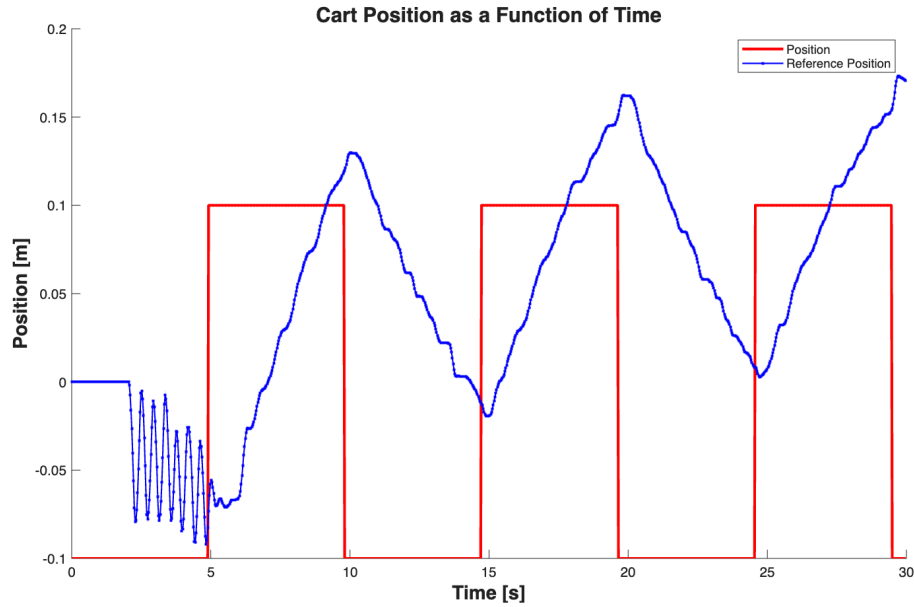


Figure 4: Experimental cart position  $z(t)$  tracking square wave reference for proportional output feedback controller (please note that the legend labels are switched) .

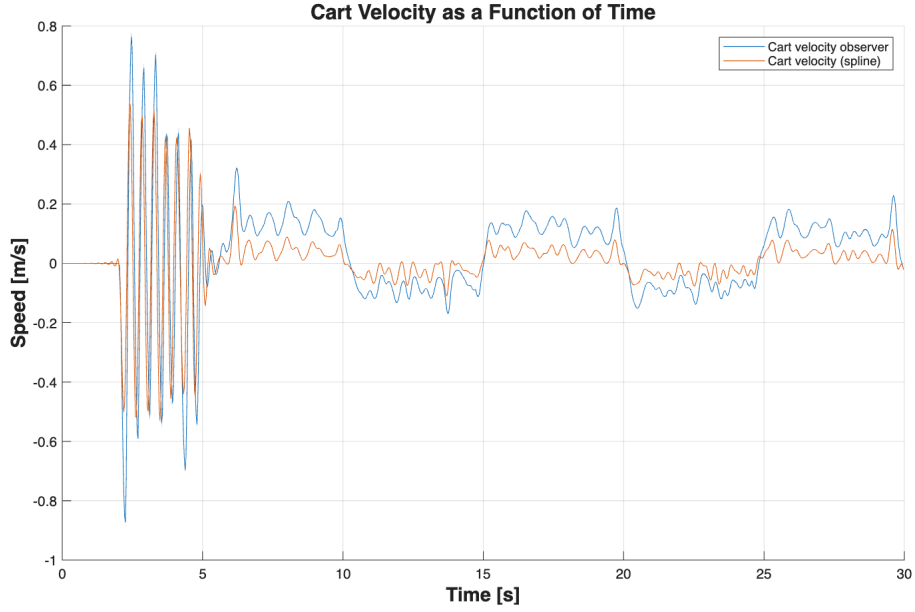


Figure 5: Experimental cart velocity (observer estimate vs. spline interpolation) for proportional output feedback controller.

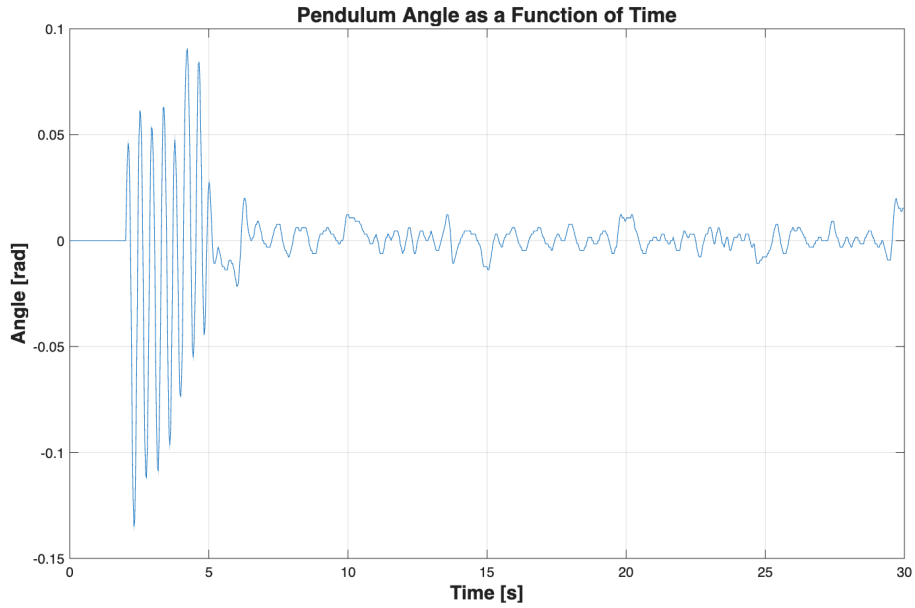


Figure 6: Experimental pendulum angle  $\theta(t)$  for proportional output feedback controller.

The experimental results demonstrate satisfactory performance of the LQR-based output feedback controller. Examining the cart position in Figure 4, the system successfully tracks the square wave reference signal after an initial transient calibration phase ( $t < 5s$ ). The measured settling time for transitions between setpoints is approximately:

**Experimental Settling Time:  $T_s \approx 3 - 4$  seconds**

The cart position  $z(t)$  successfully tracks the square wave reference signal with acceptable transient response. During the steady-state phases (between transitions at  $t = 10s, 15s, 20s, 25s$ ), the cart maintains position within a small neighborhood of the desired setpoint. However, a noticeable steady-state tracking error of approximately 0.01-0.02 m is observed, where the actual

position consistently lags the reference. This steady-state error is attributable to unmodeled friction forces in the cart rail and drive mechanism, which act as constant disturbances that the proportional controller cannot eliminate without integral action.

Figure 6 shows that the pendulum angle  $\theta(t)$  remains close to the upright equilibrium throughout the experiment, with deviations primarily occurring during square wave transitions and the initial calibration phase. After the initial transient ( $t > 5s$ ), the pendulum angle typically remains within  $|\theta| < 0.02$  rad during steady-state, validating the effectiveness of the high  $q_3 = 150$  weighting in the LQR design. The larger oscillations during transitions (reaching approximately 0.02 rad peak) are expected, as the cart must accelerate rapidly to track the step changes in reference position, temporarily disturbing the pendulum equilibrium.

Figure 5 presents an interesting comparison between the observer's velocity estimate and a spline-interpolated velocity computed from the measured position data. During the initial transient, the observer estimate exhibits larger magnitude spikes (reaching  $\pm 0.8$  m/s) compared to the spline velocity. This discrepancy arises because the observer responds to both measurement changes and model predictions, while the spline purely differentiates the noisy position signal. After convergence ( $t > 5s$ ), the two estimates track reasonably well, though the observer estimate shows slightly more oscillatory behavior, reflecting its higher bandwidth (observer poles at -40 vs. the slower system dynamics).

### 3.3 Provide a simulation plot of the signals $(z(t), \theta(t), u(t))$ and the reference signal $z_d(t)$ for the same set of controller parameters used in your experiments. Discuss the differences between simulation and experiment

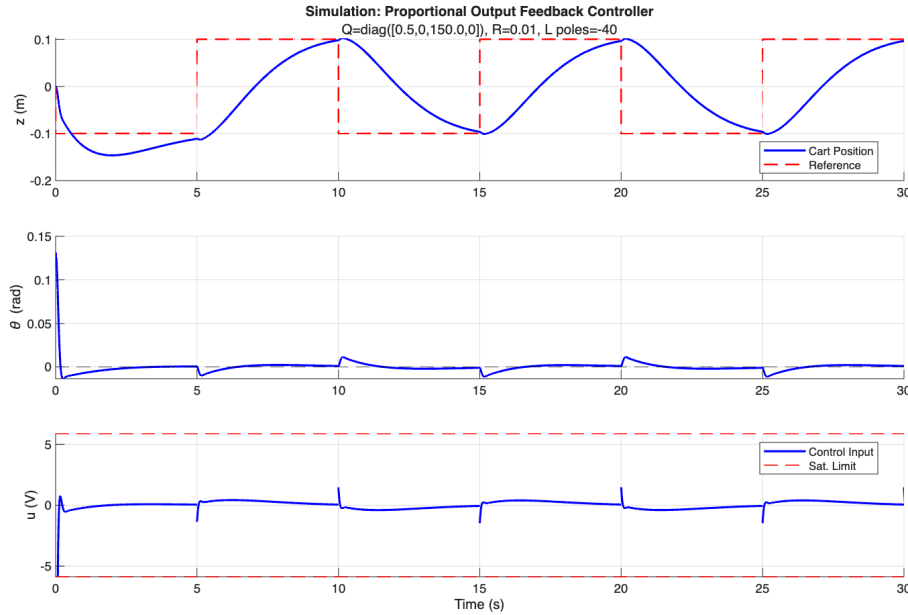


Figure 7: Experimental pendulum angle  $\theta(t)$  for proportional output feedback controller.

From the plots we notice that the position of the cart, after a transient calibration phase, attempts to track the square wave. However, in the attempt to stabilize the pendulum, the cart is not able to hold its position at either edge of the square wave. During the experiment, these effects are exaggerated. There is also a noticeable drift in the error between the tracking reference and the cart position in terms amplitude. This can be remedied by adding integral action to the observer design, which would remove the steady offset for step references. All in all, the results are expected in both the experimental and simulated plots.

## 4 5.2: Output Feedback Control with Integral Action

### 4.1 Explain the trial and error procedure that you followed to get the best performance on the experimental setup, and provide the parameters of the controllers (either the eigenvalues or the Q, R matrices) that gave the best results.

To address the steady-state tracking error observed in Section 5.1, we augmented the output feedback controller with integral action. The integral controller eliminates steady-state error by accumulating the tracking error over time, effectively rejecting constant disturbances such as friction.

The augmented controller architecture introduces an additional state  $\hat{c}$  representing the integral of the tracking error:

$$\hat{c} = \int_0^t (z_d(\tau) - z(\tau)) d\tau \quad (4)$$

The control law becomes:

$$u = K_x \hat{x} - K_c \hat{c} \quad (5)$$

where  $K_x$  is the state feedback gain and  $K_c$  is the integral gain. This structure ensures that any persistent tracking error drives  $\hat{c}$  to increase, which in turn adjusts the control input  $u$  to eliminate the error.

### Tuning Procedure

The tuning process for the integral controller followed a similar iterative approach to Section 5.1, but with the additional challenge of balancing integral action against system stability:

### Final Integral Controller Design

- **LQR Parameters:**  $Q = \text{diag}(50, 0, 30, 0)$ ,  $R = 0.2$
- **State feedback gain:**  $K_x = [15.811, 19.832, -55.196, -9.7173]$
- **Integral gain:**  $K_c = -10$
- **Observer gain:** Same  $L$  as Section 5.1 (observer design unchanged)

Notably, the integral controller uses a significantly smaller pendulum angle weight ( $q_3 = 30$ ) compared to the proportional controller ( $q_3 = 150$ ). This is because the integral action provides inherent disturbance rejection, reducing the burden on aggressive pendulum stabilization. The resulting  $K_x$  has smaller magnitudes than the proportional  $K$ , reflecting a more conservative control strategy that relies on integral action for accuracy rather than aggressive feedback for tracking.



4.2 Provide an experimental plot of the signals  $(z(t), \theta(t), u(t))$ , as well as the settling time. Discuss the results: whether they are satisfactory, whether they conform to your expectations, and so on.

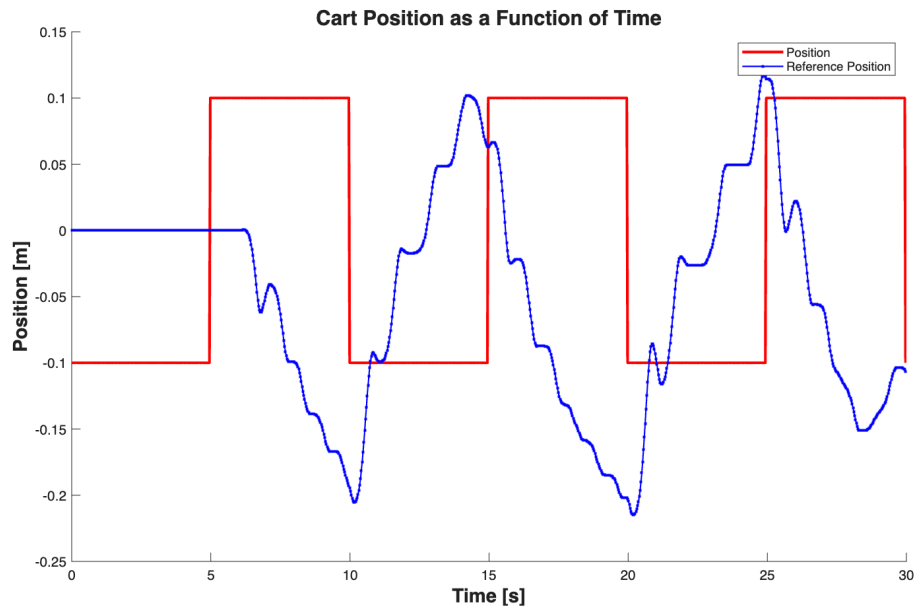


Figure 8: Experimental cart position  $z(t)$  tracking square wave reference for integral output feedback controller (please note that the legend labels are switched) .

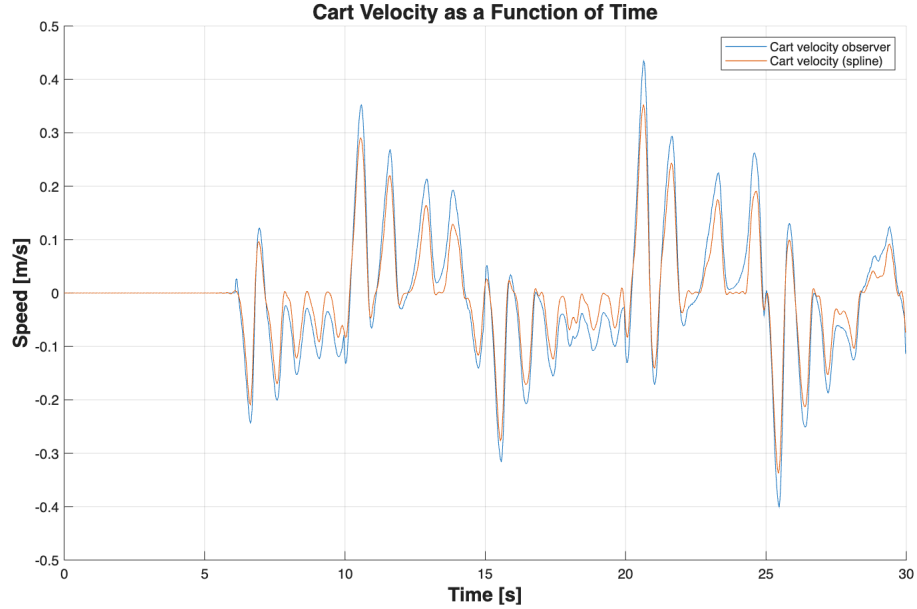


Figure 9: Experimental cart velocity (observer estimate vs. spline interpolation) for integral output feedback controller.

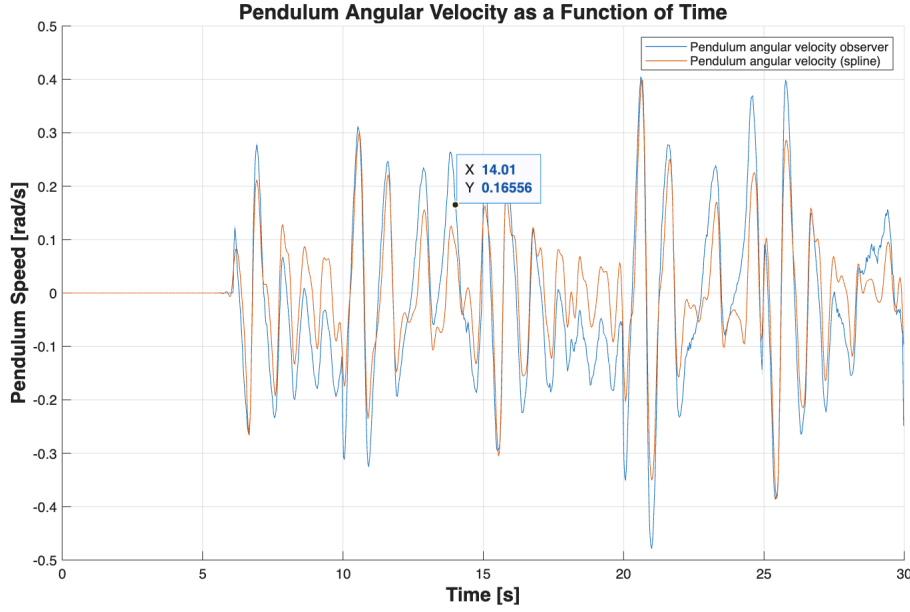


Figure 10: Experimental pendulum angle velocity  $\dot{\theta}(t)$  for integral output feedback controller.

It can be observed from the experimental plot that the tracking error in terms of steady state amplitude error is minimal. This suggests that the integral action served its purpose in reducing cumulative tracking error. However, the phase shift is still present and the cart does not fully track the square edges present in the reference signal. Since we designed the state estimation error to converge faster than the tracking error, the faster bandwidth in the observer ensures that the estimation error converges to zero more rapidly than the plant's own states are changing. This rapid convergence means the controller essentially operates with the "true" states sooner, which improves overall system performance and stability. Combined with integral action we notice lower steady state error (drift), somewhat acceptable tracking ability, and more smooth stabilization as can be observed in the cart velocity plot. While the pendulum angle data was not saved, the angular velocity plot shows that the pendulum remains well-stabilized, with velocity variations primarily occurring during square wave transitions.

#### 4.3 Compare your experimental results with the experimental results that you obtained in the previous section. Does the output feedback integral controller perform better than standard output feedback controller?

For applications requiring precise tracking in the presence of unknown constant disturbances (such as friction, gravity offsets, or manufacturing tolerances), the integral controller is the superior choice. The proportional controller may be adequate for applications where small steady-state errors are acceptable or where disturbances are negligible. In addition, the integral controller exhibited a broader and more robust set of stabilizing gain values, which made the tuning process significantly more straightforward.